# Ahmetcan Aksu

**SOFTWARE DEVELOPER — BACKEND, FINTECH & AI**

Node.js · TypeScript · C# / .NET · Rust · LLM agents & tool calling

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## SUMMARY

Software developer with 4+ years of professional experience building backend systems end to end. Fintech: at Fonmap, an SPK-licensed fund investment platform (100,000+ Google Play downloads), I single-handedly built and integrated the TEFAS order infrastructure, and I work daily with banking and capital-markets systems in C#/.NET and Node.js/TypeScript. AI: I build LLM agents with tool calling on Claude, OpenAI-compatible APIs and local open-weight models, and shipped AI-driven portfolio features to production. Underneath both: algorithms and systems work in Rust, including a type-safe programming language and bytecode VM for embedded targets, built from scratch.

## EXPERIENCE

### Backend Developer · Fonmap Financial Technologies

Feb 2024 – Present · Istanbul, Türkiye

- Led backend and frontend work from pre-production to go-live of Fonmap, a fund investment app from a Capital Markets Board (SPK)-licensed portfolio manager, live on iOS and Android with 100,000+ Google Play downloads.
- Single-handedly built and integrated the TEFAS order pipeline in .NET: order submission, status tracking, cancellation and blockage release as fault-tolerant background jobs with distributed locking and Redis caching; authored 40+ of the platform's scheduled jobs.
- Designed chain transactions: multi-step fund swaps (sell, settle, buy) that let investors rebalance a portfolio in a single request, including partial-failure and cancellation handling.
- Integrated AI-driven portfolio recommendations, fund outlooks and risk evaluations into the investment flow.
- Digitised regulated flows under Turkish capital-markets rules: venture capital and real estate fund (GSYF/GYF) issuance, qualified-investor applications and checks, KYC/AML controls, time-stamped transaction documents and account statements.
- Migrated core systems to TypeScript, then helped move the backend to C#/.NET (EF Core); moved asset delivery to a Cloudflare R2 CDN and built structured Elasticsearch logging.

**Stack:** C# / .NET, Entity Framework Core, MSSQL, Hangfire, Redis, Elasticsearch, Cloudflare R2, Node.js, TypeScript, React

### Full Stack Developer · Qpien

Aug 2022 – Feb 2024 · Istanbul, Türkiye

- Built frontend and backend for an all-in-one e-commerce platform with real-time messaging and third-party integrations (Instagram, Trendyol, Shopify).
- Engineered scalable, high-throughput systems processing 100,000+ messages per hour using asynchronous and parallel programming.
- Designed real-time communication layers on WebSockets and RabbitMQ, improving system responsiveness and reliability.
- Developed responsive interfaces in React and TypeScript, and backend APIs in Node.js and GraphQL, backed by Redis caching and MongoDB.

**Stack:** React, TypeScript, Node.js, GraphQL, WebSockets, RabbitMQ, Redis, MongoDB

### Full Stack Developer (Intern) · Massive Energy

Jun 2021 – Jul 2022 · Istanbul, Türkiye

- Independently designed and built management panels, backend architecture and databases, and managed Linux-based deployment and releases on AWS.

**Stack:** Node.js, JavaScript, MongoDB, HTML/CSS, AWS

### Freelance Developer — Mobile, Backend & Full-Stack · Self-employed

Jan 2019 – Aug 2022 · Remote

- Delivered scalable mobile and backend applications for a range of clients, owning API design, frontend–backend integration and release across web, mobile and server-side work.

**Stack:** React, Node.js, Flutter, Rust, MongoDB, JavaScript, HTML/CSS

## PROJECTS & OPEN SOURCE

### AI Sprite Studio — Creator

2026 · TypeScript · LLM agents

- Built a tool-calling LLM agent that draws and animates pixel art through 16 undoable editor tools.
- Designed palette-indexed grids, symmetry and frame deltas for token economy on local 8B models (LM Studio).

### AIStudio — AI Video Studio

2026 · TypeScript · Claude, Groq, LM Studio

- LLM agent with 20+ tools writes and edits animated HTML video frames; runs on Claude, Groq or local models.
- Deterministic export: headless Chrome under a virtual clock piped to ffmpeg, with ElevenLabs narration and subtitles.

### Ellie Language — Creator & Maintainer

2020 – Present · Rust · 48 GitHub stars

github.com/behemehal/Ellie-Language

- Designed a type-safe programming language for embedded (ARM) and sandboxed (WebAssembly) environments.
- Built the parser, compiler and bytecode VM from scratch: memory management, stack/heap allocation, garbage collection and bytecode optimisation.

### UtilStation — Full-Stack Developer

Web platform

utilstation.com

- One-stop platform bringing developer and productivity tools — file conversion, data processing, everyday utilities — together in a single app.

### Open Source — Behemehal & Personal Projects

100+ repositories · Rust · Dart · C#

github.com/behemehal · github.com/ahmetcanaksu

- Rust crates on crates.io — Menemen (streaming HTTP), Rust-NMEA (GPS parser), SafeEn (embedded database), Wole (Wake-on-LAN); CHIP-8 and 6502 emulators; Flutter apps; Dart and npm libraries.

## TECHNICAL SKILLS

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|---|---|
| **Programming** | TypeScript, JavaScript, C#, Rust, Dart, C, C++, Java, SQL |
| **AI & LLMs** | LLM agents & tool calling, Claude API, OpenAI-compatible APIs (Groq, LM Studio), local LLMs (Qwen3-VL, Gemma), vision models, token optimisation, ElevenLabs TTS |
| **Backend** | Node.js, Deno, Express, GraphQL (Apollo), REST, WebSockets, .NET, Entity Framework Core, Hangfire, RabbitMQ |
| **Frontend** | React, Redux, Svelte, Electron, Flutter, HTML/CSS, WebAssembly |
| **Databases** | MSSQL, PostgreSQL, MongoDB, Redis, Elasticsearch, MinIO |
| **Infra & Tools** | Docker, Linux, AWS, Cloudflare R2, Git, Puppeteer, CI/CD |
| **Systems & Embedded** | Algorithm design, compiler and bytecode VM design, ARM, ESP32/ESP8266, Arduino, Raspberry Pi, TCP/UDP, FFI |
| **Spoken** | Turkish — Native; English — Professional — C2 (EF SET certified) |

## EDUCATION

### BSc, Management Information Systems

Anadolu University · 2024 – Present

### Associate Degree, Computer Programming

Işık University · 2022 · GPA 3.53 / 4.00 · C, C++, C#, Java, JavaScript, MSSQL, HTML

## CERTIFICATES

EF SET English Certificate — C2; Turkish Capital Markets Association; Mobile App Development with Google Flutter; JavaScript Final Assessment — BTK Academy; C# Programming — BTK Academy; Node.js Backend Development — Patika.dev; Advanced Frontend Web Development — Patika.dev.